
3AL Tutorial 4 2024 Solutions

1. Let $s = \text{lcm}(m, n)$. Use the First Isomorphism Theorem to prove that $\mathbb{Z}_s$ is isomorphic to a subgroup of $\mathbb{Z}_m \times \mathbb{Z}_n$.

HINT: Recall that $\mathbb{Z}_n \cong \mathbb{Z}/n\mathbb{Z}$.

Notice that $\mathbb{Z}_s \cong \mathbb{Z}/s\mathbb{Z}$, similarly for $\mathbb{Z}_m$ and $\mathbb{Z}_n$. We will construct a homomorphism φ from $\mathbb{Z}$ to $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ with kernel $s\mathbb{Z}$. Then $\mathbb{Z}_s \cong \mathbb{Z}/s\mathbb{Z}$ will be isomorphic to a subgroup, $\varphi(\mathbb{Z})$, of $\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$.

The most natural map $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ would be $\varphi(x) = (x(m\mathbb{Z}), x(n\mathbb{Z}))$. Now, for $x, y \in \mathbb{Z}$, we have

$$\begin{aligned}\varphi(x + y) &= ((x + y)m\mathbb{Z}, (x + y)n\mathbb{Z}) \\ &= (xm\mathbb{Z} + ym\mathbb{Z}, xn\mathbb{Z} + yn\mathbb{Z}) \\ &= (xm\mathbb{Z}, xn\mathbb{Z}) + (ym\mathbb{Z}, yn\mathbb{Z}) \\ &= \varphi(x) + \varphi(y).\end{aligned}$$

So φ is a homomorphism. It suffices to show that φ is a homomorphism since we only need to show that $\mathbb{Z}/s\mathbb{Z}$ is isomorphic to a subgroup of $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$, not the whole group. Basically, we only need to use the first version of the First Isomorphism Theorem.

Now, $x \in \ker \varphi$ if and only if $\varphi(x) = (xm\mathbb{Z}, xn\mathbb{Z}) = (m\mathbb{Z}, n\mathbb{Z})$. That is, $x \in m\mathbb{Z}$ and $x \in n\mathbb{Z}$. However, $x \in m\mathbb{Z}$ if and only if x is a multiple of m . Similarly, $x \in n\mathbb{Z}$ if and only if x is a multiple of n . Thus $x \in \ker \varphi$ if and only if x is a multiple of both m and n . The set of all such multiples can be expressed as $s\mathbb{Z}$, where $s = \text{lcm}(m, n)$ since all multiples of m and n are multiples of s . Thus $\ker \varphi = s\mathbb{Z}$.

By the First Isomorphism Theorem, $\mathbb{Z}_s \cong \mathbb{Z}/s\mathbb{Z} \cong \varphi(\mathbb{Z})$ where $\varphi(\mathbb{Z})$ is a subgroup of $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$. Since $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_m \times \mathbb{Z}_n$, there is an isomorphism φ' between the two groups. Then $\varphi'\varphi(\mathbb{Z})$ is a subgroup of $\mathbb{Z}_m \times \mathbb{Z}_n$ and so $\mathbb{Z}_s$ is isomorphic to $\varphi'\varphi(\mathbb{Z})$ which is a subgroup of $\mathbb{Z}_m \times \mathbb{Z}_n$.

2. Let H and K be normal subgroups of a group G such that $G = HK$.

(a) Prove that $G/(H \cap K) \cong (G/H) \times (G/K)$.

HINT: Use the First Isomorphism Theorem. To prove surjectivity, note that for any $g_1, g_2 \in G$, we have $g_1^{-1}g_2 \in G = HK$. So $g_1^{-1}g_2 = hk$ for some $h \in H$ and $k \in K$. Consider the element $x = g_2k^{-1} = g_1h \in G$.

(b) Prove that if H and K are disjoint, then $G/K \cong H$ and $G/H \cong K$.

NOTE: Two subgroups are said to be disjoint if their intersection is the trivial subgroup $\{1\}$.

NOTE: We will use these results later in the course.

(a) Let's construct an epimorphism $\varphi : G \rightarrow (G/H) \times (G/K)$. Recall that the elements of G/H and G/K are cosets, so they have the form gH and gK for $g \in G$. Therefore, the map defined by $\varphi(g) = (gH, gK)$ is quite a natural choice for φ . We now need to show that φ is an epimorphism and $\ker \varphi = H \cap K$.

Take $g, g' \in G$, then

$$\begin{aligned}\varphi(gg') &= (gg'H, gg'K) = ((gH)(g'H), (gK)(g'K)) \\ &= (gH, gK)(g'H, g'K) = \varphi(g)\varphi(g').\end{aligned}$$

So, φ is a homomorphism.

To prove that φ is an epimorphism, let (g_1H, g_2K) be an arbitrary element of $(G/H) \times (G/K)$. Notice that the representatives of the two cosets are not necessarily the same! Since $G = HK$, there exist $h \in H$ and $k \in K$ such that $g_1^{-1}g_2 = hk$. Thus $g_2k^{-1} = g_1h$. Let this element be denoted by x . Then

$$\varphi(x) = (xH, xK) = (g_1hH, g_2k^{-1}K) = (g_1H, g_2K).$$

Thus φ is an epimorphism.

Now, notice that the identity of $(G/H) \times (G/K)$ is (H, K) (as H is the identity of G/H and K is the identity of G/K). An element $g \in G$ belongs to $\ker \varphi$ if and only if $\varphi(g) = (H, K)$. That is, $gH = H$ and $gK = K$. But this holds precisely when $g \in H$ and $g \in K$, meaning that $g \in H \cap K$. Thus $\ker \varphi = H \cap K$.

Then, by the First Isomorphism Theorem, $G/(H \cap K) \cong (G/H) \times (G/K)$.

(b) By the Second Isomorphism Theorem, we have

$$H/(H \cap K) \cong HK/K.$$

We are given that $G = HK$. If H and K are disjoint then $H \cap K = \{1\}$. So $H/(H \cap K) = H/\{1\} \cong H$. Moreover, $HK = G$. So we have $H \cong G/K$. Similarly, $K \cong G/H$.

3. Let K be a normal subgroup of a finite group G such that K has prime index p in G . Prove that for all subgroups H of G either:

- (i) $H \subseteq K$, or,
- (ii) $G = HK$ and $|H : H \cap K| = p$.

Let H be a subgroup of G . By the Second Isomorphism Theorem, HK is a subgroup of G with $K \triangleleft HK$. Note that HK/K is a subgroup of G/K . Therefore, $|HK : K|$ divides $|G : K| = p$. Thus either $|HK : K| = 1$ or p . If $|HK : K| = 1$, then $K = HK$ and so $H \subseteq K$. That is, (i) holds. On the other hand, if $|HK : K| = p$, then

$$\frac{|G|}{|K|} = |G : K| = p = |HK : K| = \frac{|HK|}{|K|}$$

and so $|G| = |HK|$ and thus $G = HK$. Moreover, by the Second Isomorphism Theorem,

$$|H : H \cap K| = \frac{|H|}{|H \cap K|} = \frac{|HK|}{|K|} = |HK : K| = p.$$

This means (ii) holds.

4. Let G be a group with normal subgroups H and K such that $H \subseteq K$. Prove that if G/H is cyclic, then so are G/K and K/H .

Since G/H is cyclic, so are its subgroups. Thus, K/H is cyclic. Moreover, since G/H is cyclic, so are its factor groups. By the Third Isomorphism Theorem, $G/K \cong (G/H)/(K/H)$. Thus $(G/H)/(K/H)$ is cyclic and then so is G/K , since it is isomorphic to the aforementioned group.

5. Let a and b be positive integers such that $a \mid b$. Determine the order of the factor group $a\mathbb{Z}/b\mathbb{Z}$.

Note that since $a \mid b$, we have $b\mathbb{Z} \subseteq a\mathbb{Z} \subseteq \mathbb{Z}$. Moreover, since $\mathbb{Z}$ is abelian, these subgroups are normal. By the Third Isomorphism Theorem,

$$(\mathbb{Z}/b\mathbb{Z})/(a\mathbb{Z}/b\mathbb{Z}) \cong \mathbb{Z}/a\mathbb{Z}.$$

However, $\mathbb{Z}/a\mathbb{Z} \cong \mathbb{Z}_a$ and so $|\mathbb{Z}/a\mathbb{Z}| = |\mathbb{Z}_a| = a$. Similarly, $|\mathbb{Z}/b\mathbb{Z}| = b$. By Lagrange's Theorem,

$$|(\mathbb{Z}/b\mathbb{Z})/(a\mathbb{Z}/b\mathbb{Z})| = \frac{|\mathbb{Z}/b\mathbb{Z}|}{|a\mathbb{Z}/b\mathbb{Z}|} = \frac{b}{|a\mathbb{Z}/b\mathbb{Z}|}.$$

However, due to the Third Isomorphism Theorem, this must equal $|\mathbb{Z}/a\mathbb{Z}| = a$. Thus

$$a = \frac{b}{|a\mathbb{Z}/b\mathbb{Z}|} \implies |a\mathbb{Z}/b\mathbb{Z}| = \frac{b}{a}.$$

Extra problems

Recall that for subgroup H and K of a group, we have the product HK . In an additive group, we typically write the product as $H + K = \{h + k : h \in H, k \in K\}$.

Consider the additive group $\mathbb{Z}$ and let $a, b \in \mathbb{Z}^+$.

- (I) Prove that $a\mathbb{Z} + b\mathbb{Z} = \gcd(a, b)\mathbb{Z}$.

HINT: For all $a, b \in \mathbb{Z}^+$, there exist $n, m \in \mathbb{Z}$ such that $\gcd(a, b) = an + bm$.

- (II) Prove that $a\mathbb{Z} \cap b\mathbb{Z} = \text{lcm}(a, b)\mathbb{Z}$.

(III) Use the Second Isomorphism Theorem to show that $ab = \gcd(a, b) \operatorname{lcm}(a, b)$.

- (I) We have $a\mathbb{Z} \subseteq \gcd(a, b)\mathbb{Z}$ and $b\mathbb{Z} \subseteq \gcd(a, b)\mathbb{Z}$. Then $a\mathbb{Z} + b\mathbb{Z} \subseteq \gcd(a, b)\mathbb{Z}$, since $\gcd(a, b)\mathbb{Z}$ is a subgroup of $\mathbb{Z}$. For $\gcd(a, b)z \in \gcd(a, b)\mathbb{Z}$, we can write $\gcd(a, b)z = (an + bm)z = a(nz) + b(mz) \in a\mathbb{Z} + b\mathbb{Z}$. Thus $\gcd(a, b)\mathbb{Z} \subseteq a\mathbb{Z} + b\mathbb{Z}$, so the two sets are equal.
- (II) Note that $z \in a\mathbb{Z} \cap b\mathbb{Z}$ if and only if $a \mid z$ and $b \mid z$. That is, $\operatorname{lcm}(a, b) \mid z$ and so $z \in \operatorname{lcm}(a, b)\mathbb{Z}$.
- (III) Note that $a\mathbb{Z}$ and $b\mathbb{Z}$ are both normal subgroups of $\mathbb{Z}$. Then, by the Second Isomorphism Theorem,

$$a\mathbb{Z}/(a\mathbb{Z} \cap b\mathbb{Z}) \cong (a\mathbb{Z} + b\mathbb{Z})/b\mathbb{Z}.$$

Using (a) and (b), we can write this as

$$a\mathbb{Z}/\operatorname{lcm}(a, b)\mathbb{Z} \cong \gcd(a, b)\mathbb{Z}/b\mathbb{Z}.$$

Since the groups are isomorphic, their orders must be equal. Notice that $a \mid \operatorname{lcm}(a, b)$ and $\gcd(a, b) \mid b$. We use Question 5 to then get $|a\mathbb{Z}/\operatorname{lcm}(a, b)\mathbb{Z}| = \operatorname{lcm}(a, b)/a$ and $|\gcd(a, b)\mathbb{Z}/b\mathbb{Z}| = b/\gcd(a, b)$. Thus, $\operatorname{lcm}(a, b)/a = b/\gcd(a, b)$ and so $ab = \gcd(a, b) \operatorname{lcm}(a, b)$, as desired.